

Topic: Section 1

Place your figure of the IMU flipping results. Add a brief caption (2-3 sentences) that explains how the post hoc digital filtering we performed here is different from the real-time IMU data filtering from Lab 3. Include one pro and one con of each approach, in your own words.

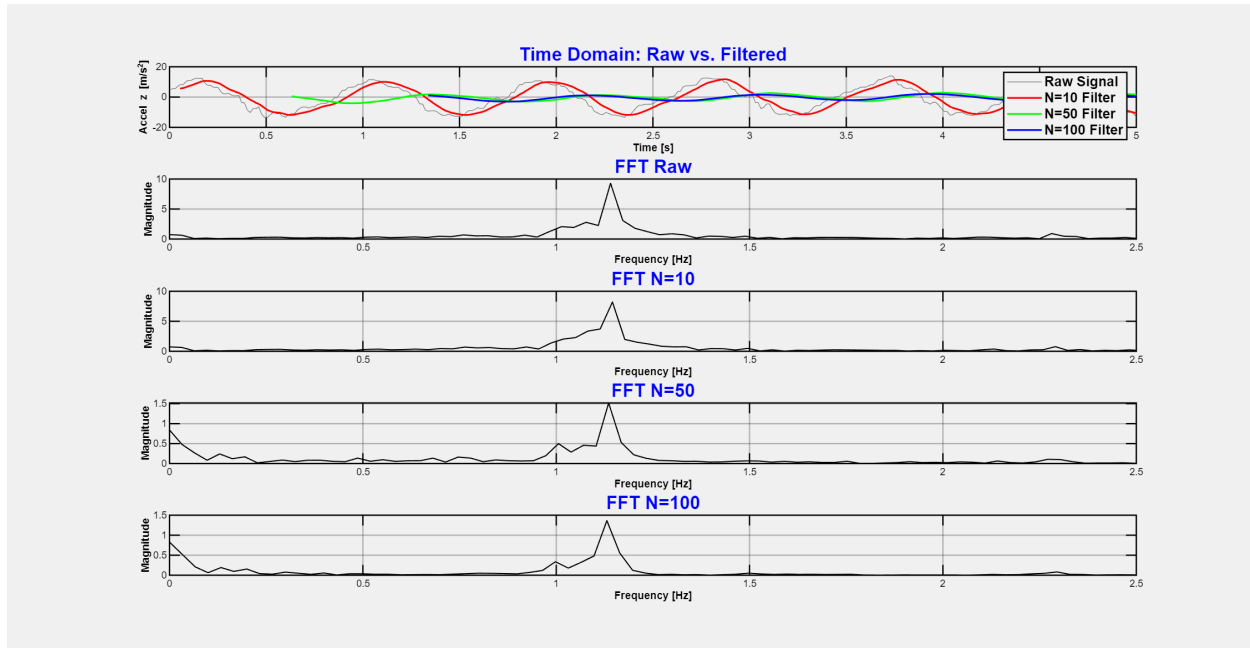


Figure 1: IMU flipping Zoomed in

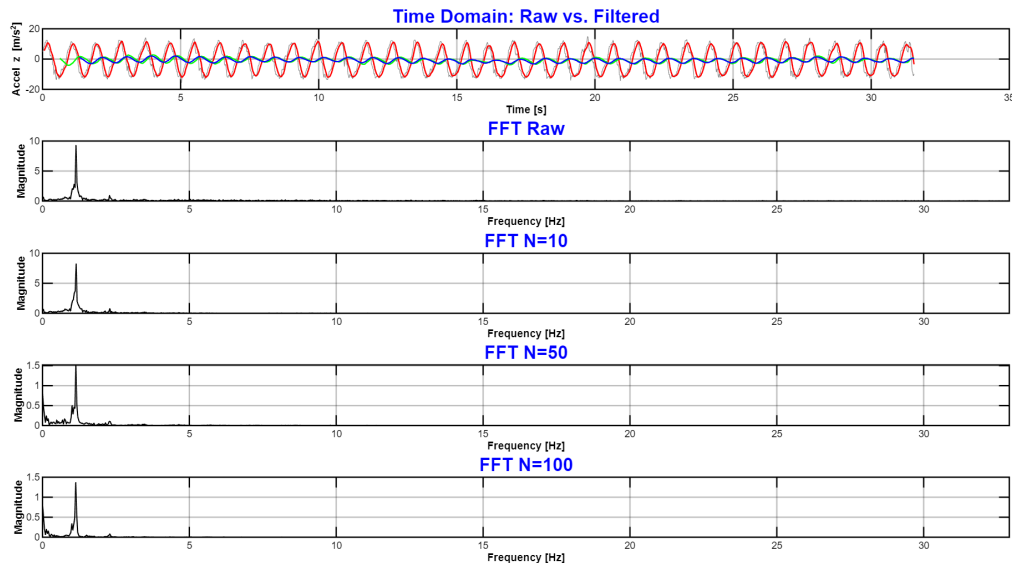


Figure 2: IMU flipping

Well they were 2 different situations. Lab 3 was having us do data filtering on the arduino

and Lab 7 had us do all the data filtering with MATLAB on our larger computers. In all, the arduino had us limited to really basic filters like butters, and moving average. Using MATLAB lets us use the larger FFT and the auxiliry filters. I still like how steady the arduino was, even if it was a little messy. Its like Little Ceasers, Hot and Ready.

Topic: Section 2

Place your figure of the four music frequency response spectrums. Add a brief caption (2-3 sentences) about the effect, both in the frequency response and audibly, of the filtering and downsampling on the music.

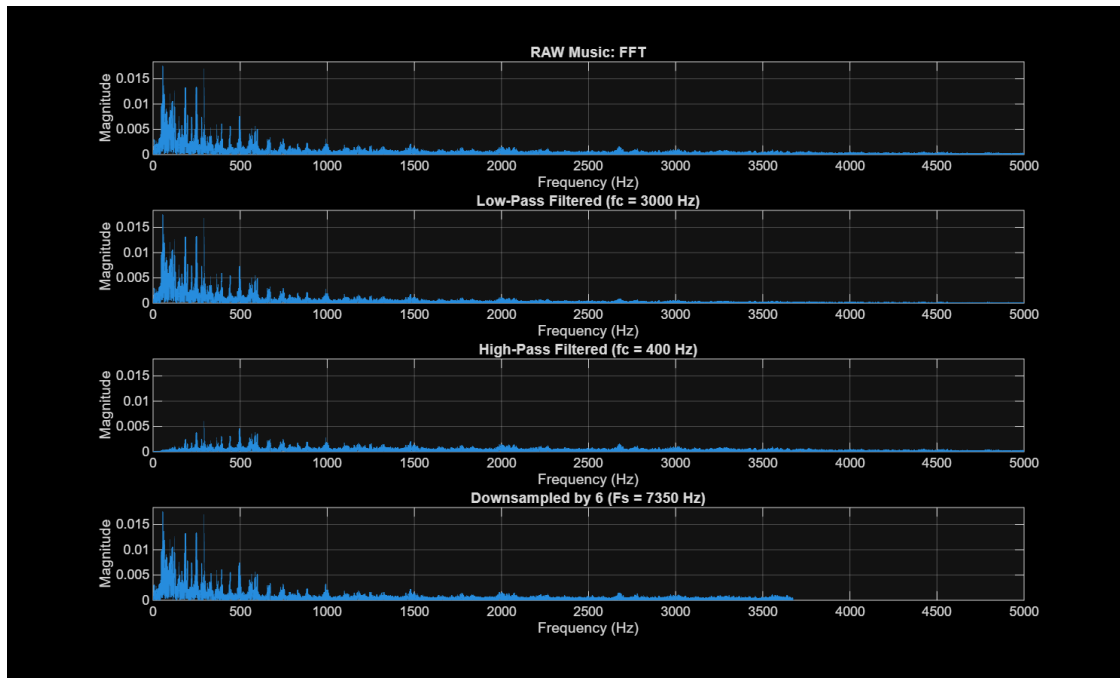


Figure 3: low pass, high pass, band pass and downsampled

The effect of the filters was pretty "linear" in that it did what I expected. The down-sampling taking the already built $1 \times N$ matrix and skipping every 6 spaces. It feels like its further away and underwater a little. The LP filter definitely gave a high ended product and the HP gave the opposite. It makes sense since we are cutting those out.

Topic: Section 3

Place your figure of the joke and/or laughter audio data FFT. Briefly (in 1-2 sentences) explain your approach for isolating the meaningful part of the signal. If it helps for your explanation, you may want to annotate the figure.

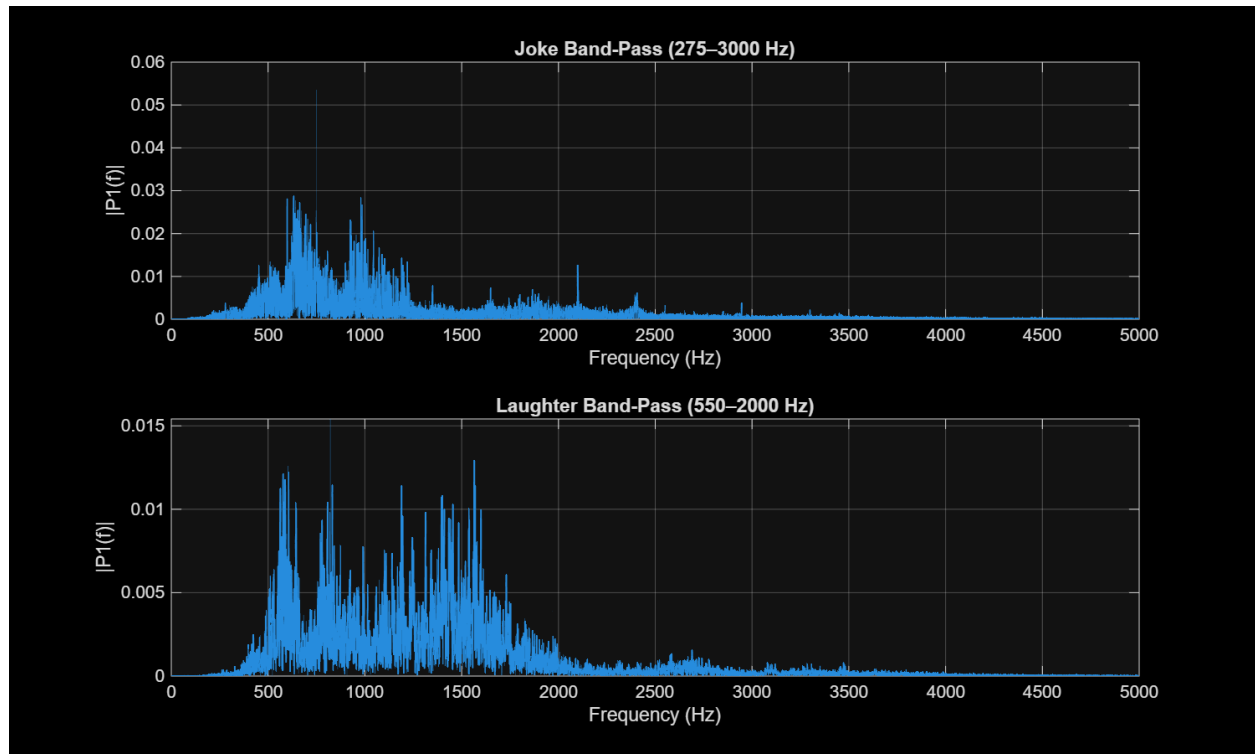


Figure 4: joke

Geez, this part was a little difficult since I pretty much was just letting the google results of "voice frequency" and "laughter frequency" guide my guess and check.

Topic: Section 4

Please list the following: 1. Any online sources you used and what they helped with. If you used any generative AI tools (e.g., ChatGPT, Gemini, Copilot, Claude, etc.), include a screenshot of your entire chatlog. 2. Who you worked with in the lab (excluding teachers) and what they helped with. Online sources should be notated cleanly with the name and a hyperlink. “What they helped with” should focus on conciseness. Examples: “Wikipedia helped me understand citations,” “John worked with me on Sign-off 2.” If the answer is “No online sources used.” or “No collaborators.”, please note this in the report for the corresponding item, so we know you considered both prompts.

List online sources and collaborators.

<https://www.mathworks.com/help/matlab/ref/stackedplot.html>

<https://www.mathworks.com/help/matlab/ref/fft.html>

<https://www.youtube.com/watch?v=2kvemW3W1Vk>

https://www.mathworks.com/help/matlab/ref/for.html?sid=srchtitle_support_results1_for+loops

<https://www.mathworks.com/help/signal/smoothing-and-denoising.html>

<https://www.mathworks.com/matlabcentral/answers/2153300-finding-peaks-in-a-very-noisy-signal>

I worked with Madison Flanders.

Page 5: Self-evaluation:

- (a) What did you expect to learn? I was really hoping to be able to use the FFT and filters with this lab. Im glad that it had that.
- (b) How would you rate your attendance and participation in class during this lab experience? Are any adjustments needed? My attendance was trash this week. Im feeling the end of term pressure already. Why is school hard?
- (c) How well did you do at the lab assignment itself? You might include thinking about strengths and weaknesses of your performance in this response. Please also give yourself a grade (out of 4 possible points). I would give myself a 4 for this. I kept on top of everything. Outside of my attendance of course.
- (d) What did you actually learn in the lab? (This might include a range of things, from subject matter and concrete skills to ways of knowing and working together with peers.) I got more MATLAB work in and that always a really fun thing to do now that I have gotten over the apprehension. Learning how to use the filter1() function is pretty great.

Code

%CODE HERE
